



Aggie Reuse Store: WQ26 Donor Research Q's

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Introduction:

The following document is a follow-up to the Tableau Dashboard analysis. Rather than extracting information from visuals, the subsequent inferences were obtained using statistical testing methods. The report focuses on answering three questions proposed at the start of developing this project:

- 1) Is there an association between donor type (student, staff, community) and their motivations to donate?
- 2) Is there a correlation between visits to ARS and the number of items donated?
- 3) Is there a difference in the number of student years and the number of items donated?

The scheduled fourth question was ultimately omitted due to the high number of NULL values and data quality issues.

Development:

Introduction:

The Donor Analysis Project was built on two core tasks: one being a Tableau Storyboard, the other proposing/answering hypothesis questions. The project was designed so that both visual and statistical analyses could be performed, maximizing the use of available data.

Therefore, once the Tableau Development was concluded, the next step was to begin building the script used to answer our research questions. This process would begin in early March and conclude later in the month, developed in Jupyter Notebook(Python).

Why Jupyter Notebook?

Similar to R-Studio, Jupyter Notebook is a powerful development tool that allows users to run a select chunk of code, enabling users to plot multiple visuals, run different tests, and rerun select portions of data in a neat, document-style format. This is perfect for our multi-layer process, as readability and replaying code (due to user errors) is prioritized over a single, multilayered script.

Non-Parametric Tests

Due to the rather small and skewed I sample size (~50 subjects), conducting normal-approximated tests may lead to inaccurate results. Thankfully, Non-parametric tests offer more robustness than their restrictive counterpart, being distribution-free and less sensitive to skewedness. This causes Non-Parametrics to be perfectly suited for our dataset, with analysis being conducted using:

- Chi-Squared Test (Independence)
- Spearman's Rank Correlation
- Kruskal-Wallis H Test

Data Processing:

Similar to the Dashboard Analysis, there are data cleaning steps that need to be run before we can proceed with testing. Because the dataset was cleaned before running Tableau analysis, the following steps were taken to transform our data, exploding/ordinally scaling information.

String Splitting:

One of the questions in the survey asks a donor for their reasons for donating. This question would later be used to see if it affected the motives of a donor, dictating if there was an association between donors and motives.

However, the data collection methods led to donation reasons being compiled under one roof, with reasons not being split across multiple columns (i.e, [Charity, Declutter] instead of a column for each). As a result, our test will not run properly unless we successfully split each cause. Thus, string splitting was employed. All entries were split and duplicated by the number of reasons a user selected (if a user selected 2 reasons, they now have two rows). This allows us to run the test successfully, even if there is some overlap in our data.

```
String Splitting "Donation Reason"

df["Donation Reason"].dtype
dtype('O')

df_q1 = df.copy()

df_q1["Donation Reason"] = df_q1["Donation Reason"].astype(str)

# Splits all reasons by a ","
df_q1["Donation Reason"] = df_q1["Donation Reason"].str.split(",")

# Explodes the list into individual rows specifically for the Q1 dataframe
df_q1 = df_q1.explode("Donation Reason")

# Removes white space
df_q1["Donation Reason"] = df_q1["Donation Reason"].str.strip()
```

Figure 1: Code snippet of String Splitting.

Enumerating “Frequency of Visits”

Additionally, donors were asked how frequently they visited ARS. Responses were measured in temporal statements such as “once a week”, or “once a quarter. This question would be used to see if there is a possible correlation between visits and the number of items donated.

While the following descriptions are great for interpretation, numeric tests such as Spearman’s Rank Correlation only accept numeric entries. To achieve this, we will enumerate the frequency of visits from 1-6, 1 being “Never Visited” and 6 being “More

than once a week.” By doing so, our once categorical variable can now be measured on a numeric scale, allowing us to perform numeric comparisons.

```
Enumerating "Frequency of Visits"

#Maps each value scaled on visit frequency.
visit_mapping = {
    "More than once a week": 6,
    "About once a week": 5,
    "About once per month": 4,
    "About once per quarter": 3,
    "Less than once a quarter": 2,
    "I have never visited": 1
}

#Update Visit col.
df["Visits"] = df["Visit Frequency"].map(visit_mapping)
df = df.dropna(subset=["Visits"]).copy()
df["Visits"]
```

Figure 2: Code snippet of Enumerating String Variables.

Analysis:

The analysis portion of our report will uncover both the reasons for our proposed questions and the results of testing. This testing process attempts to aid the Aggie Reuse Store in future decisions. All tests will be run with a confidence level of 95% ($\alpha = 0.05$) to ensure accurate answers.

Association Between Donors and Motives

With many donors coming through and out of our organization, based on one's affiliation, do different people donate for different reasons? Answering this question assists ARS in comprehending its donors' motives and exploring whether they varies between demographics.

After running the test, we were given a p-value of 0.99. Given how close the value is to 1, there is likely a confounding variable/bias present.

Upon further investigation, the cause of this problem is the sample sizes. While Students have a sample size of 90, our staff counts and community numbers are far lower (20, 8). With this knowledge, we'll take the conclusion with a grain of salt. Our test results state there is not enough statistically significant evidence of an association between a type of donor and their motives.

```
#pulling specific statistics:
chi2, p, dof, expected = chi2_contingency(cont_table)

print("Chi-square:", chi2) #test stat
print("p-value:", p)

Chi-square: 3.562058080808081
p-value: 0.9901074723534397
```

Figure 3: Code snippet of Chi-Squared Results

Correlation Between Number of Visits and Items Donated

With hundreds of visitors every week, likely, many donors also shop at ARS. This question ponders the likelihood of donors offering more based on the times they visit ARS. If there is a correlation, ARS could prioritize efforts in retaining its consumers, as doing so could lead to abundant future donations.

Surprisingly, our test results show an incredibly small p-value (<1%), with $\rho = -0.44$. This is shocking because a negative ρ value indicates an inverse relationship. Therefore, the more someone visits, the less they are likely to donate. Given the p-value <1%, our test results indicate statistically significant evidence that the more someone visits ARS, the less they donate.

```
rho, p = spearmanr(df["Visits"],
                   df["Total Items"])

print("Spearman rho:", rho) #test stat
print("p-value:", p)

Spearman rho: -0.4443937299026817
p-value: 0.0022263235991933383
```

Figure 4: Code snippet of Spearman's Correlation.

Difference in Class and Items Donated

Located in the Memorial Union of UC Davis, ARS's largest demographic is students. The final question wonders if a student will donate more, given their school level. Answering the following may help ARS in appealing to upperclassmen or the inverse if true.

Our final test outputs a p-value of 0.24. This result means there is not enough statistically significant evidence to prove a relationship between student years and the number of items they donate.

```
#grouping total items by school year:
groups = [
    group["Total Items"]
    for name, group in df.groupby("School Year")
]

stat, p = kruskal(*groups)

print("H statistic:", stat) #state stat
print("p-value:", p)

H statistic: 6.756893629043683
p-value: 0.23935730009037248
```

Figure 5: Code snippet of Kruskal-Wallis H-Test.

Conclusion:

The Donor Analysis Project was built on two parts: a storyboard and three research questions. This report answered the latter; the results may not be entirely accurate, given small sample sizes and other quirks.

1. There is no association between the type of donor (student, staff, community) and their motives for donating.
 - a. **Limitation:** Student sample size was much larger (90) than both staff and community members (8,20)
2. The more someone visits ARS, the less they are likely to donate.
 - a. Shows ARS could appeal to donors who have never visited before
 - b. Many donors only want to donate, rather than gain more clothing.
3. There is no relationship between class and items donated.
 - a. Student level is not a significant measure of how much someone donates.